

DSA1101

Introduction to Data Science

Week 13 Tutorial 11

TODAY'S AGENDA

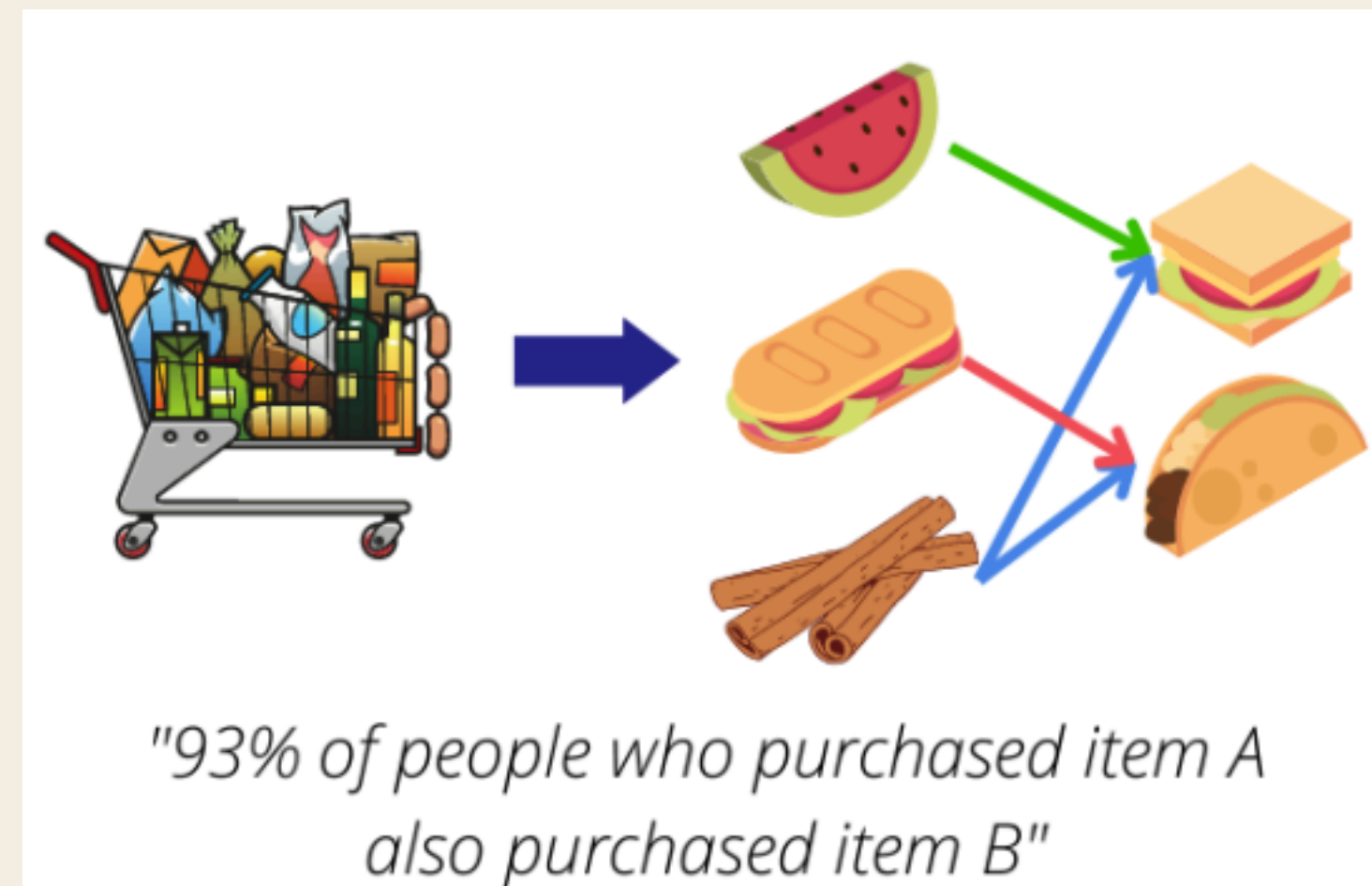
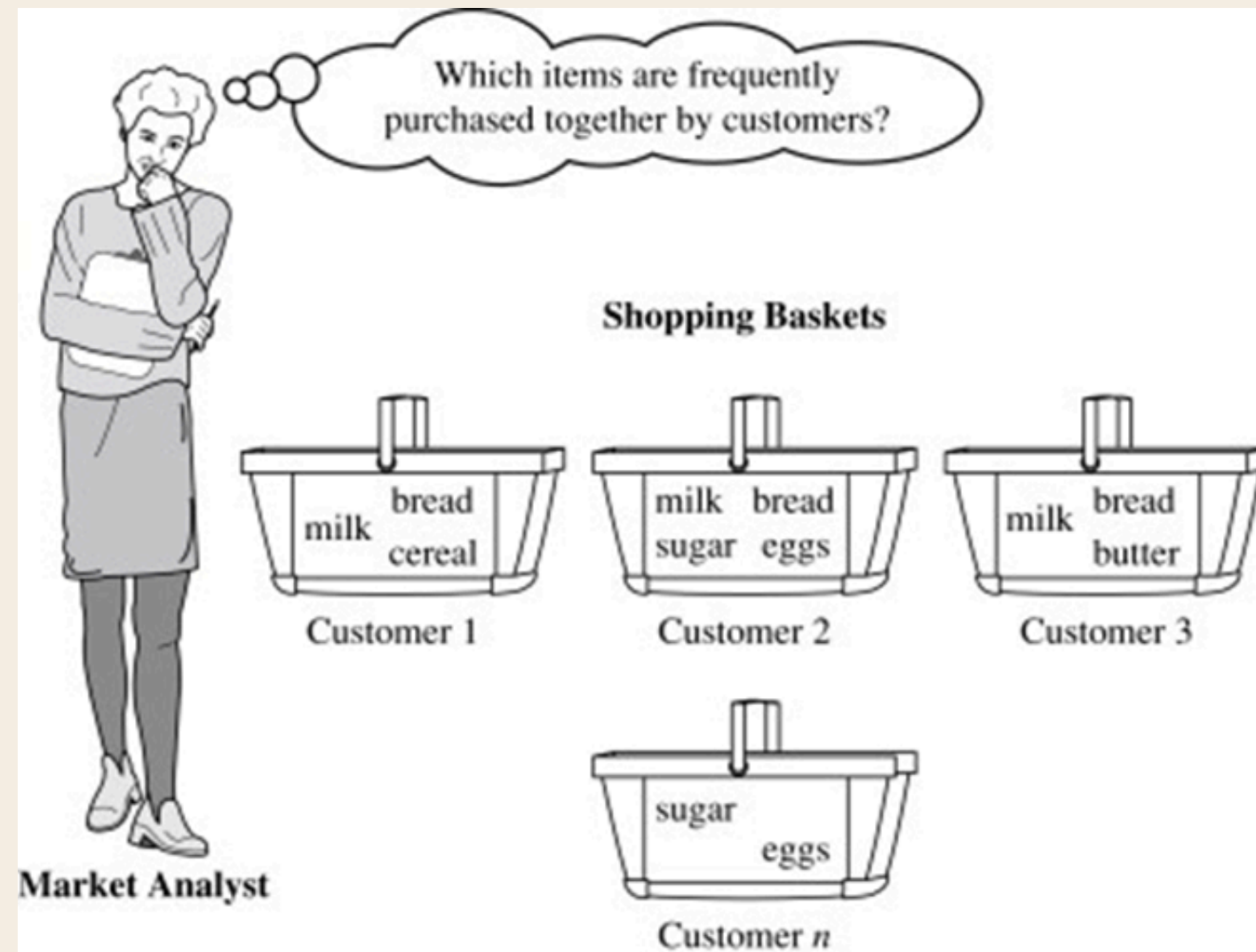
1. Quick Revision
2. Off-site Questions Discussion
3. Say goodbye :(



QUICK REVISION

ASSOCIATION RULES

Unsupervised machine learning technique used to discover interesting relationships or "if-then" association rules between variables in large datasets



ITEMSETS

Itemset: collection of items or individual entities

k-itemset: itemset containing k items {item 1, item2, ..., item k}

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

2-itemset

4-itemset

4-itemset

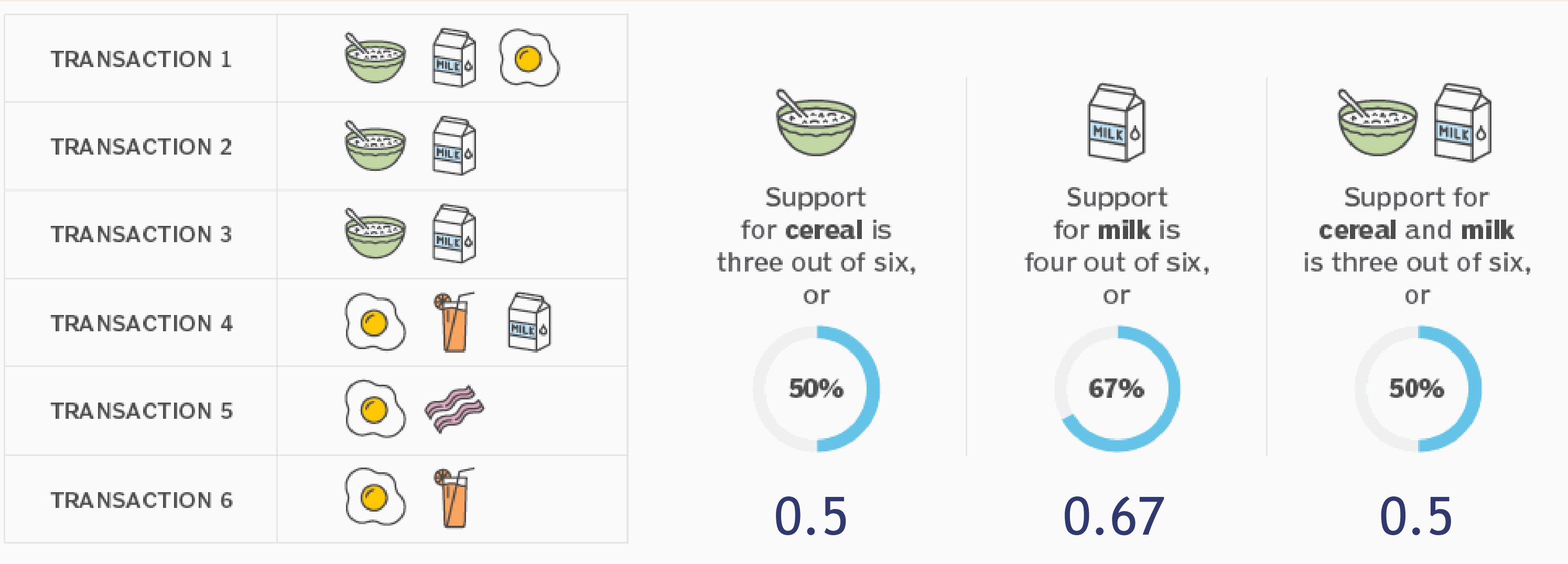
4-itemset

4-itemset

SUPPORT OF AN ITEMSET

Support of Itemset {L} = $\text{count}(\text{transactions containing L}) / \text{total transactions}$

eg. Say a dataset with 100 transactions.if 80 transactions contain itemset {bread} then support of itemset {bread} is 0.8

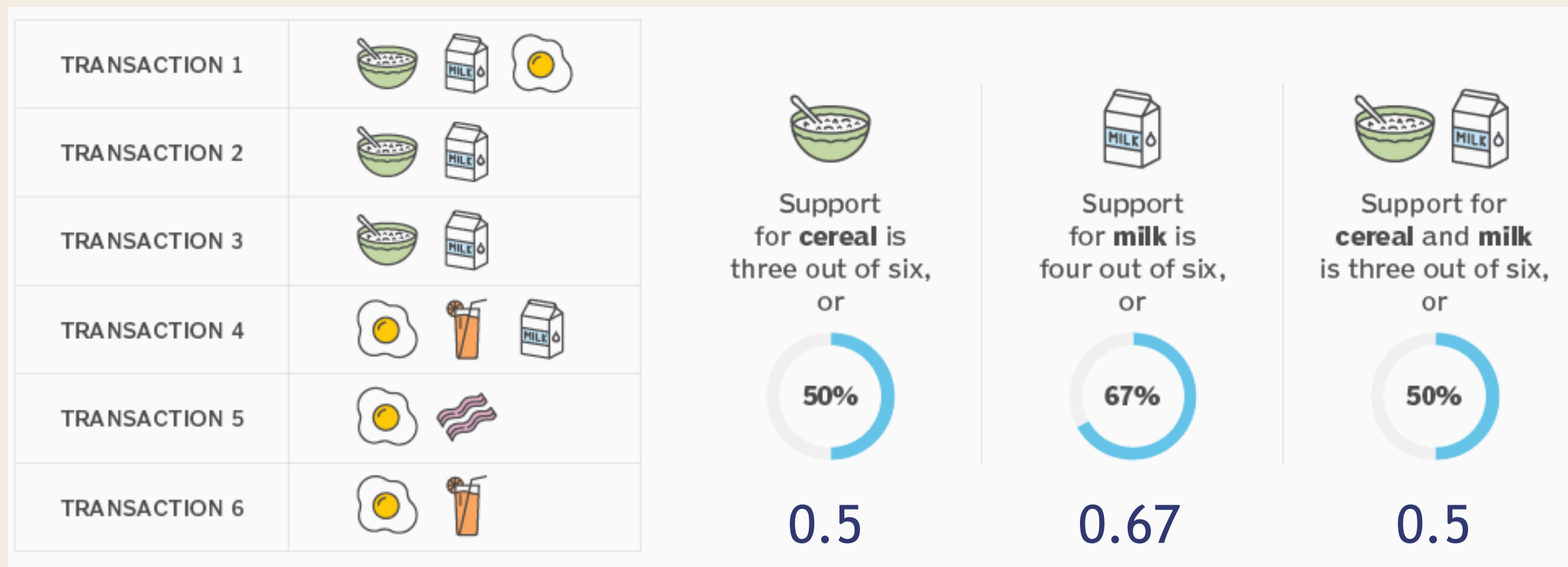


MINIMUM SUPPORT

Frequent itemset: Support of itemset $\geq$ Minimum Support (defined by you!)

Apriori property: If an itemset is frequent, then all of its subsets must also be frequent.

Eg. If {cereal, milk} appears in 50% of transactions, then {cereal} alone and {milk} alone must each appear in at least 50% of transactions too.



So if minimum support is 0.5, then {cereal, milk} is **frequent**, and hence {cereal} and {milk} alone is also frequent.

APRIORI ALGORITHM (bottom-up initiative in your notes)

Minimum support = 0.6

Transaction ID	Items Bought
T1	Bread, Butter, Milk
T2	Bread, Butter
T3	Bread, Milk
T4	Butter, Milk
T5	Bread, Milk

Repeat this process until:
1. No frequent itemsets found at the current size
2. Maximum itemset size (iteration) allowed

Output:
All frequent itemsets: {bread}, {butter}, {milk}, {bread, milk}

These itemsets are candidates for generating and evaluating association rules.

1. Determine all possible 1-itemset and their support
 $\{\text{bread}\} \rightarrow 0.8, \{\text{butter}\} \rightarrow 0.6, \{\text{milk}\} \rightarrow 0.8$
2. Keep all “frequent” 1-itemset that meet minimum support threshold
 $\{\text{bread}\} \rightarrow 0.8, \{\text{butter}\} \rightarrow 0.6, \{\text{milk}\} \rightarrow 0.8$
3. Pair all “frequent” 1-itemsets into 2-itemsets and get their support
 $\{\text{bread, butter}\} \rightarrow 0.4, \{\text{bread, milk}\} \rightarrow 0.6, \{\text{butter, milk}\} \rightarrow 0.4$
4. Keep all “frequent” 2-itemset that meet minimum support threshold
 $\{\text{bread, milk}\} \rightarrow 0.6$

GENERATING AND EVALUATING RULES

Rules are formed from frequent itemsets that have 2 or more items

Our example before: {bread}, {butter}, {milk}, **{bread, milk}**

Create rules by splitting them to LHS and RHS:

{bread} → {milk} and {milk} → {bread}

Each rule captures a buying pattern: {bread} → {milk} means 'people who buy bread tend to also buy milk'. We evaluate how strong and reliable each rule is using **confidence, lift, and leverage**.

$$Confidence(X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X)}$$

$$Lift(X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X) \times Support(Y)}.$$

$$Leverage(X \rightarrow Y) = Support(X \wedge Y) - Support(X) \times Support(Y)$$

CONFIDENCE

Measure of **certainty** and **trustworthiness** associated with each rule

$$Confidence(X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X)}$$

Since denominator is always bigger or equals to numerator (based on the Apriori property) **Confidence** ≤ 1

e.g if {bread, egg, milk} has support 0.15 and {bread, egg} also 0.15 then

confidence {bread, egg} $\rightarrow$ {milk} is 1

(100% of time customer buys bread and egg, milk is bought as well

rule is correct for 100% transaction containing {bread, egg})

PROBLEM WITH CONFIDENCE

Only takes Support(X) into account in denominator and not Support(Y)

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

Suppose in 1000 transactions:

- X appears in 200 transactions
- X and Y appears tgt in 150 transactions
- Y appears in 950 transactions

$$\text{Confidence}(X \rightarrow Y) = 150 / 200 = \mathbf{0.75}$$

Confidence is quite high showing people who buy X often buys Y too. Is this a meaningful rule then?

Not really, because Y is popular itself alr (Support = 0.95).

This rule presents the false idea of X driving people to buy Y.
But in reality people were going to buy Y anyway.

In fact buying X actually reduces the chance of Y from 95% down to 75%.

Let's see how Lift solves the problem!

LIFT

How much more likely two items are to be purchased together compared to if they were independent of each other.

$$Lift(X \rightarrow Y) = \frac{Support(X \wedge Y)}{Support(X) \times Support(Y)}.$$

- If lift = 1 $\rightarrow$ independent
- If lift > 1 $\rightarrow$ they co-occur more than independence would predict
- If lift < 1 $\rightarrow$ they co-occur less than independence would predict

Back to our example from previous slide:

$$Confidence(X \rightarrow Y) = 0.75$$

$$Lift(X \rightarrow Y) = 0.15 / (0.2 \times 0.95) = 0.789$$

Lift < 1, meaning buying X actually makes buying Y less likely, even though confidence is quite high!

A quick example of how this happens in irl:

{butter} $\rightarrow$ {margarine}.

People who buy butter rarely also buy margarine (as they are substitutes) so Lift likely <1

LEVERAGE

Similar to Lift, but instead of ratio uses differences

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \wedge Y) - \text{Support}(X) \times \text{Support}(Y)$$

- If leverage = 0 $\rightarrow$ independent
- If leverage > 0 $\rightarrow$ they co-occur more than independence would predict
- If leverage < 0 $\rightarrow$ they co-occur less than independence would predict

But why do we need leverage when we have lift already?

Rule	Supp($X \wedge Y$)	Supp(X)	Supp(Y)	Lift	Leverage
$A \rightarrow B$	0.40	0.50	0.50	1.6	0.15
$C \rightarrow D$	0.004	0.005	0.005	160	0.003

Small Support inflates Lift. $C \rightarrow D$ has a lift of 160 which looks like it has HUGE effect, but leverage of only 0.003 reveals it barely has any in reality.

APRIORI ALGORITHM IN R

Package:

```
#install.packages("arules")  
#install.packages("arulesViz")
```

```
library(arules)  
library(arulesViz)
```

Load Data:

The data is a 'transactions' object in the arules package

```
data(Groceries)  
?Groceries # Description of data
```

Groceries {arules}

R Documentation

The Groceries Transactions Data Set

Description

The `Groceries` data set contains 1 month (30 days) of real-world point-of-sale transaction data from a typical local grocery outlet. The data set contains 9835 transactions and the items are aggregated to 169 categories.

Format

Object of class [transactions](#).

Inspect and Explore Data:

```
inspect(head(Groceries))
```

```
> inspect(head(Groceries))  
items  
[1] {citrus fruit,  
    semi-finished bread,  
    margarine,  
    ready soups}  
[2] {tropical fruit,  
    yogurt,  
    coffee}  
[3] {whole milk}
```

```
summary(Groceries)
```

```
transactions as itemMatrix in sparse format with  
9835 rows (elements/itemsets/transactions) and  
169 columns (items) and a density of 0.02609146
```

most frequent items:

whole milk	other vegetables	rolls/buns	soda
2513	1903	1809	1715
yogurt	(Other)		
1372	34055		

element (itemset/transaction) length distribution:
sizes

1	2	3	4	5	6	7	8	9
2159	1643	1299	1005	855	645	545	438	350
10	11	12	13	14	15	16	17	18
246	182	117	78	77	55	46	29	14
19	20	21	22	23	24	26	27	28
14	9	11	4	6	1	1	1	1
29	32							
3	1							

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	3.000	4.409	6.000	32.000

APRIORI ALGORITHM IN R

Inspect and Explore Data:

```
# Display all 169 grocery labels  
Groceries@itemInfo[1:10,]
```

```
> Groceries@itemInfo[1:10,]  
      labels level2      level1  
1 frankfurter sausage meat and sausage  
2 sausage sausage meat and sausage  
3 liver loaf sausage meat and sausage  
4 ham sausage meat and sausage  
5 meat sausage meat and sausage  
6 finished products sausage meat and sausage  
7 organic sausage sausage meat and sausage  
8 chicken poultry meat and sausage  
9 turkey poultry meat and sausage  
10 pork pork meat and sausage
```

```
# Indicates which item labels appear in every transaction  
Groceries@data[1:10,]
```

```
> Groceries@data[1:10,]  
10 x 9835 sparse Matrix of class "ngCMatrix"  
  
[1,] . . . . . | . . . . .  
[2,] . . . . . | . . . . .  
[3,] . . . . .  
[4,] . . . . .  
[5,] . . . . .  
[6,] . . . . .  
[7,] . . . . .  
[8,] . . . . . | . . . . .  
[9,] . . . . .  
[10,] . . . . . | . . . . .
```


APRIORI ALGORITHM IN R

Common Question: Find number of **1-itemsets** that are frequent when the **minimum support is 0.005**. Report the most 5 frequent 1-itemsets.

```
itemsets.1 = apriori(data,  
                      parameter = list(minlen = 1, maxlen = 1, # only 1-itemsets  
                                       support = 0.005, target="frequent itemsets"))  
  
summary(itemsets.1)  
  
inspect(head(sort(itemsets.1, by = "support"), 5))
```

```
> summary(itemsets.1)  
set of 59 itemsets
```

```
> inspect(head(sort(itemsets.1, by = "support"), 5))  
      items      support  count  
[1] {whole milk} 0.2555160 2513  
[2] {other vegetables} 0.1934926 1903  
[3] {rolls/buns} 0.1839349 1809  
[4] {soda} 0.1743772 1715  
[5] {yogurt} 0.1395018 1372
```

If question asks to do the same but for 2-itemsets or 3-itemsets,
just update minlen and maxlen!

APRIORI ALGORITHM IN R

You can also do **target = “rules”** and can **sort by confidence, lift, leverage** too! (since it's rules)

```
rules_1 <- apriori(Groceries,  
                   parameter = list(minlen=2, maxlen=4,  
                                     support=0.001, target="rules"))  
  
summary(rules_1)  
  
inspect(head(sort(rules_1, by = "lift"), 5))
```

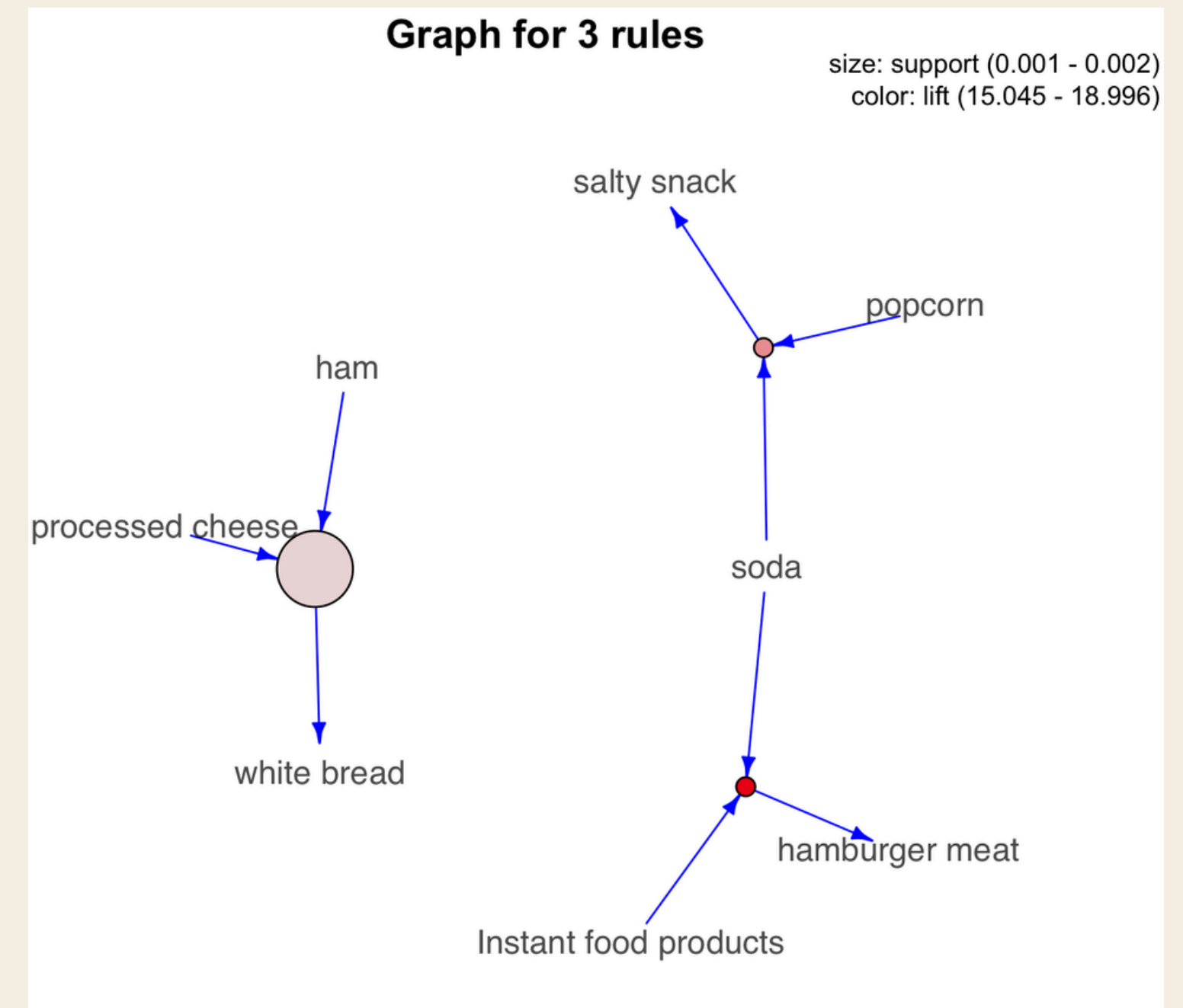
```
> inspect(head(sort(rules.1, by = "lift"), 5))
```

lhs	rhs	support	confidence	coverage	lift	count
[1] {liquor, red/blush wine}	=> {bottled beer}	0.001931876	0.9047619	0.002135231	11.235269	19
[2] {citrus fruit, grapes, fruit/vegetable juice}	=> {tropical fruit}	0.001118454	0.8461538	0.001321810	8.063879	11
[3] {root vegetables, butter, cream cheese }	=> {yogurt}	0.001016777	0.9090909	0.001118454	6.516698	10
[4] {sausage, pip fruit, sliced cheese}	=> {yogurt}	0.001220132	0.8571429	0.001423488	6.144315	12
[5] {tropical fruit, butter, white bread}	=> {yogurt}	0.001118454	0.8461538	0.001321810	6.065542	11

APRIORI ALGORITHM IN R

Common Question 2: Get the **association rules** using support = 0.001 and confidence = 0.6; and plot a figure to visualize the **top 3 rules with highest lift**.

```
rules <- apriori(Groceries,  
  parameter=list(support = 0.001,confidence = 0.6,  
  target = "rules"))  
  
highLiftRules <- head(sort(rules, by = "lift"), 3)  
  
plot(highLiftRules, method = "graph", engine = "igraph",  
  edgeCol = "blue", alpha = 1)
```



OFF-SITE QUESTIONS 1,2

Today we demonstrate
this using slides

OFF-SITE QUESTION 1A

1. A local retailer has a database that stores 10,000 transactions of last summer. After analyzing the data, a data science team has identified the following statistics:

{battery} appears in 6000 transactions $6000/10000 = 0.6$

{sunscreen} appears in 5000 transactions 0.5

{sandals} appears in 4000 transactions 0.4

{bowls} appears in 2000 transactions 0.2

{battery, sunscreen} appears in 1500 transactions 0.15

{battery, sandals} appears in 1000 transactions 0.1

{battery, bowls} appears in 250 transactions 0.025

{battery, sunscreen, sandals} appears in 600 transactions 0.06

Recall:

Support of Itemset $\{L\}$ =
 $\text{count}(\text{transactions containing } L) / \text{total transactions}$

Question 1A: What are the support values of the preceding itemsets?

OFF-SITE QUESTION 1B

1. A local retailer has a database that stores 10,000 transactions of last summer. After analyzing the data, a data science team has identified the following statistics:

{battery} appears in 6000 transactions $6000/10000 = 0.6$

{sunscreen} appears in 5000 transactions 0.5

{sandals} appears in 4000 transactions 0.4

{bowls} appears in 2000 transactions 0.2

{battery, sunscreen} appears in 1500 transactions 0.15

{battery, sandals} appears in 1000 transactions 0.1

{battery, bowls} appears in 250 transactions 0.025

{battery, sunscreen, sandals} appears in 600 transactions 0.06

**Ans: All except
{battery, bowls} are
frequent.**

**Question 1B: Assuming the minimum support is 0.05, which itemsets
are considered frequent?**

OFF-SITE QUESTION 1C

Q: What are the confidence values of {battery} → {sunscreen} and {battery, sunscreen} → {sandals}? Which of these two rules is more “interesting”?

{battery} appears in 6000 transactions 0.6
{sunscreen} appears in 5000 transactions 0.5
{sandals} appears in 4000 transactions 0.4
{bowls} appears in 2000 transactions 0.2
{battery, sunscreen} appears in 1500 transactions 0.15
{battery, sandals} appears in 1000 transactions 0.1
{battery, bowls} appears in 250 transactions 0.025
{battery, sunscreen, sandals} appears in 600 transactions 0.06

“Interesting” = Higher confidence.
Hence {battery, sunscreen} → {sandals}

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

Confidence of {battery} → {sunscreen}.

$$\frac{\text{Supp}(\{\text{battery, sunscreen}\})}{\text{Supp}(\{\text{battery}\})} = 0.15/0.6 = 0.25.$$

Confidence of {battery, sunscreen} → {sandals}.

$$\frac{\text{Supp}(\{\text{battery, sunscreen, sandals}\})}{\text{Supp}(\{\text{battery, sunscreen}\})} = 0.06/0.15 = 0.4.$$

OFF-SITE QUESTION 2A

Suppose for three products A , B and C , $\text{support}(\{A\}) = 0.6$, $\text{support}(\{B\}) = 0.6$, $\text{confidence}(\{B\} \rightarrow \{A\}) = 0.9$ and $\text{confidence}(\{C\} \rightarrow \{A, B\}) = 0.5$. Compute the following quantities.

- (a) $\text{Lift}(\{A\} \rightarrow \{B\})$
- (b) $\text{Leverage}(\{A\} \rightarrow \{B\})$
- (c) $\text{Confidence}(\{A\} \rightarrow \{B\})$
- (d) $\text{Lift}(\{A, B\} \rightarrow \{C\})$

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \wedge Y) - \text{Support}(X) \times \text{Support}(Y)$$

$$\text{Lift}(\{A\} \rightarrow \{B\}) = \frac{\text{Supp}(\{A \wedge B\})}{\text{Supp}(\{A\}) \text{Supp}(\{B\})} = \frac{\text{Confidence}(\{B\} \rightarrow \{A\}) \times \text{Supp}(\{B\})}{\text{Supp}(\{A\}) \text{Supp}(\{B\})} = \frac{(0.9)(0.6)}{(0.6)(0.6)} = 1.5$$

Ans: 1.5

OFF-SITE QUESTION 2B

Suppose for three products A , B and C , $\text{support}(\{A\}) = 0.6$, $\text{support}(\{B\}) = 0.6$, $\text{confidence}(\{B\} \rightarrow \{A\}) = 0.9$ and $\text{confidence}(\{C\} \rightarrow \{A, B\}) = 0.5$. Compute the following quantities.

- (a) $\text{Lift}(\{A\} \rightarrow \{B\})$
- (b) $\text{Leverage}(\{A\} \rightarrow \{B\})$
- (c) $\text{Confidence}(\{A\} \rightarrow \{B\})$
- (d) $\text{Lift}(\{A, B\} \rightarrow \{C\})$

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X) \times \text{Support}(Y)}.$$

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \wedge Y) - \text{Support}(X) \times \text{Support}(Y)$$

$$\begin{aligned}\text{Leverage}(\{A\} \rightarrow \{B\}) &= \text{Supp}(\{A \wedge B\}) - \text{Supp}(\{A\}) \times \text{Supp}(\{B\}) \\ &= (0.9)(0.6) - (0.6)(0.6) = 0.18\end{aligned}$$

Ans: 0.18

OFF-SITE QUESTION 2C

Suppose for three products A , B and C , $\text{support}(\{A\}) = 0.6$, $\text{support}(\{B\}) = 0.6$, $\text{confidence}(\{B\} \rightarrow \{A\}) = 0.9$ and $\text{confidence}(\{C\} \rightarrow \{A, B\}) = 0.5$. Compute the following quantities.

- (a) $\text{Lift}(\{A\} \rightarrow \{B\})$
- (b) $\text{Leverage}(\{A\} \rightarrow \{B\})$
- (c) $\text{Confidence}(\{A\} \rightarrow \{B\})$
- (d) $\text{Lift}(\{A, B\} \rightarrow \{C\})$

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \wedge Y) - \text{Support}(X) \times \text{Support}(Y)$$

$$\text{Confidence}(\{A\} \rightarrow \{B\}) = \frac{\text{Supp}(\{A \wedge B\})}{\text{Supp}(\{A\})} = \frac{0.54}{0.6} = 0.9$$

Ans: 0.9

OFF-SITE QUESTION 2D

Suppose for three products A , B and C , $\text{support}(\{A\}) = 0.6$, $\text{support}(\{B\}) = 0.6$, $\text{confidence}(\{B\} \rightarrow \{A\}) = 0.9$ and $\text{confidence}(\{C\} \rightarrow \{A, B\}) = 0.5$. Compute the following quantities.

- (a) $\text{Lift}(\{A\} \rightarrow \{B\})$
- (b) $\text{Leverage}(\{A\} \rightarrow \{B\})$
- (c) $\text{Confidence}(\{A\} \rightarrow \{B\})$
- (d) $\text{Lift}(\{A, B\} \rightarrow \{C\})$

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \wedge Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \wedge Y) - \text{Support}(X) \times \text{Support}(Y)$$

$$\begin{aligned} \text{Lift}(\{A, B\} \rightarrow \{C\}) &= \frac{\text{Supp}(\{A, B \wedge C\})}{\text{Supp}(\{A \wedge B\}) \times \text{Supp}(\{C\})} \\ &= \frac{\text{Confidence}(\{C\} \rightarrow \{A, B\}) \times \text{Supp}(\{C\})}{\text{Supp}(\{A \wedge B\}) \times \text{Supp}(\{C\})} \\ &= \frac{0.5}{0.54} = 0.93 \end{aligned}$$

Ans: 0.93

OFF-SITE QUESTION 3

Let's go to code!

FINAL PAST EXAM PAPER

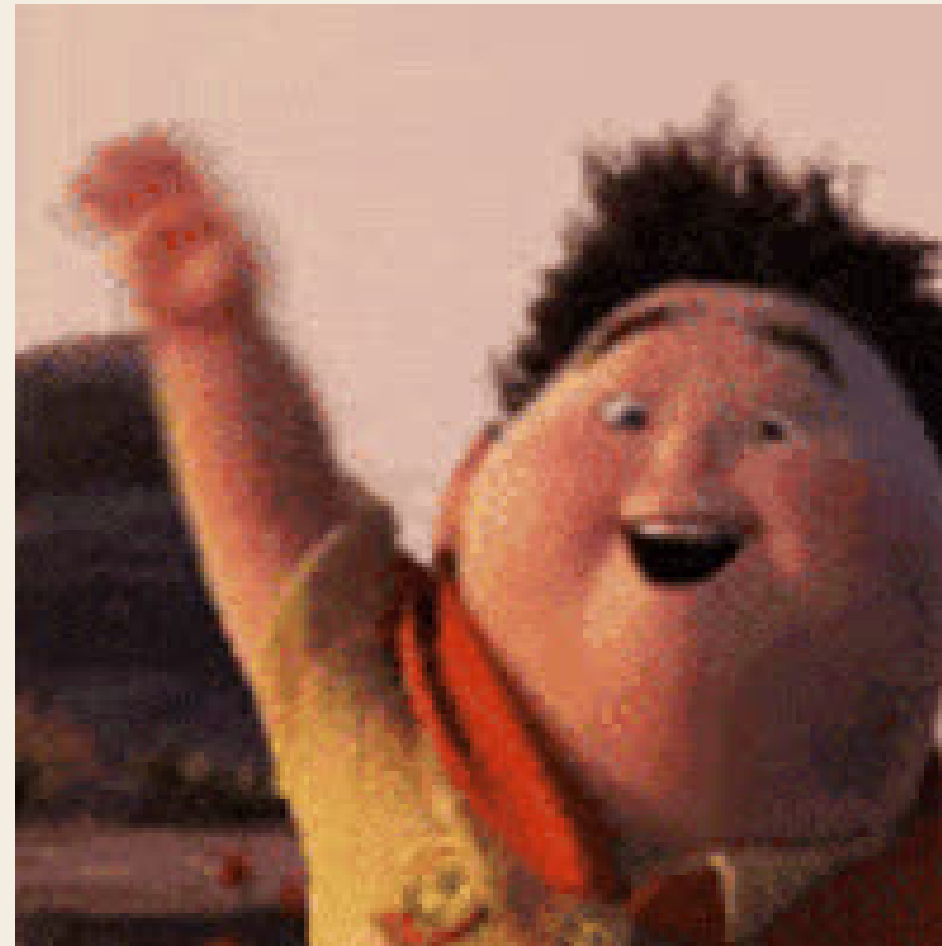


<https://drive.google.com/drive/folders/1C1IH9wrNe2Gobv0D5xuP97xr5IA27NMv?usp=sharing>

THANK YOU SO MUCH!

All the best for your final exam(s)! And see you
around school :D

Please take some chocolate
from the bag!



If you enjoyed my
teaching, it would be
great if you could
nominate me for
teaching award



Don't hesitate to tele me if you have any questions!